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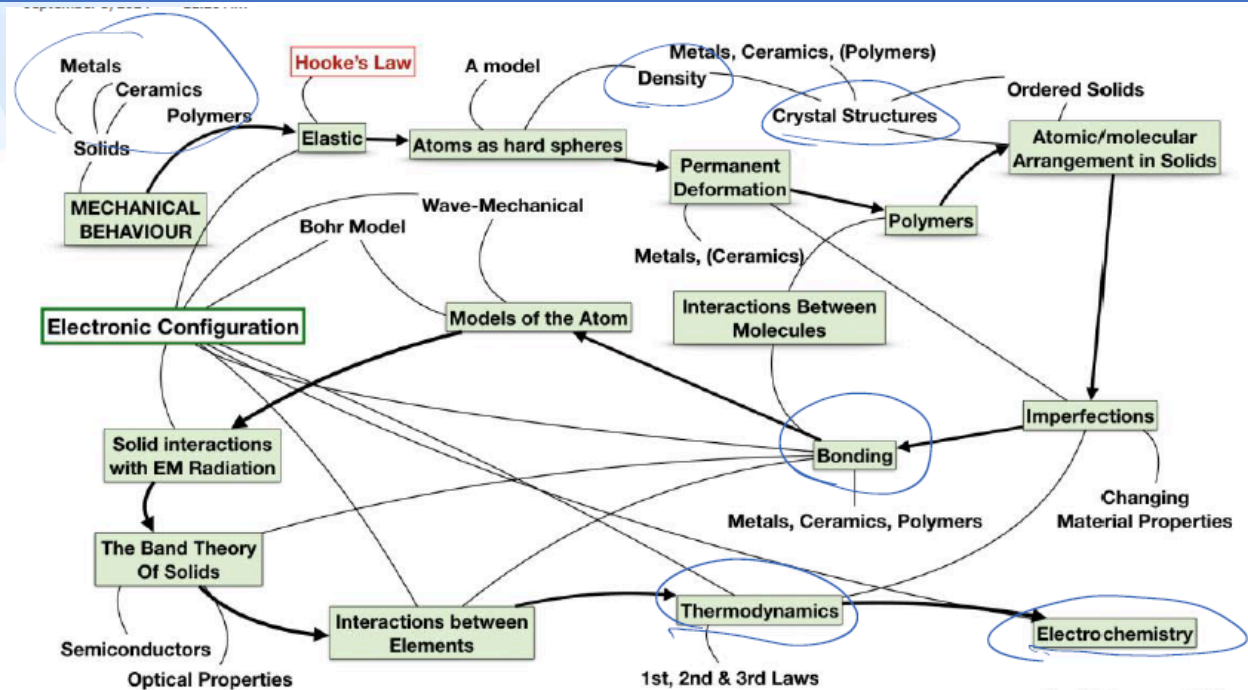
Class 公开课

课程导师

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课程内容涵盖



必备词汇表

Metal: 金属

Ceramics: 陶瓷

Polymers: 聚合物

Mechanical Behaviour: 机械特性, 指材料在外部力的作用下的变形, 破坏等现象

Elastic: 弹性

Plastic: 塑性

Permanent Deformation: 永久变形: 物体在受到外力作用后, 形状或结构发生不可逆转的改变。

Electron Configuration: 电子构型

Bonding: 键合

Radiation: 辐射

Band Theory: 能带理论

Thermodynamics: 热力学

Electrochemistry: 电化学

今日课程目标(LEARNING OBJECTIVE)

CHAPTER 1 INTRODUCTION TO MATERIAL WORLD

1. Name the Three Major Material Classes材料学三大主要类别
(Metals金属, Ceramics陶瓷, and Polymers高分子聚合物)
2. Give examples of major class materials
3. Elaborate on the shortcoming缺点 of this classification scheme -Disadvantage
4. Defend the benefit优势 of this classification scheme -Advantage

Comparison of Material Classes

Property	Metals 🏠	Ceramics 🍷	Polymers 🌿
Bonding			
Appearance			
Density			
Strength			
Electrical Conductivity			
Thermal Conductivity			
Malleability & Ductility			
Melting Point			
Examples			
Applications			

知识点讲解

□ 知识点 1: Metal(金属)

Metals(金属) are one of the three broad classes of materials (Metals, Ceramics, and Polymers).

They have distinctive physical and chemical properties that make them very useful.



EXAMPLES OF METALS & APPLICATIONS

- Iron (Fe)铁:
 - construction, machinery 常用于建筑、机械制造材料
- Copper (Cu)铜:
 - electrical wiring 常用于电气布线材料
- Aluminum (Al)铝:
 - lightweight materials, cans, aircraft 常用于轻质材料、罐子、飞机材料
- Titanium (Ti):
 - medical implants, aerospace 常用于医疗植入物、航空航天(坚固且耐腐蚀)材料

GENERAL CHARACTERISTICS OF METALS

- **Appearance 外观**
 - **Luster(Shiny) 光泽外观**
 - Metals reflect light, giving them a shiny, polished look
- **Malleability and Ductility 可塑性和延展性**
 - Malleable: Can be hammered or rolled into sheets 可以锤打或碾压成薄片状
 - Ductile: Can be drawn into wires 能够被拉成细丝
- **Density and Strength 密度和强度**
 - Most metals are high dense and mechanically strong
- **Metallic Bonding 金属键 -> Good Conductivity 传导性**
 - Highly Organized-Crystalline 在原子层面上高度有序的晶体结构
 - “sea of electrons” 金属原子之间的电子云共享形成的
- **Electrical Conductivity 导电性:**
 - Conduct electricity well
- **Thermal Conductivity 导热性:**
 - Metals transfer heat efficiently.
- **Melting and Boiling Points 熔点和沸点**
 - Typically high (though exceptions like mercury exist)



知识点2: Ceramics(陶瓷)

Ceramics(陶器) are **inorganic, non-metallic materials**(无机非金属材料) made from compounds of metals and nonmetals (like oxides, nitrides, and carbides). They are usually formed by heating加热 (firing烧制的方式) and then cooling冷却



EXAMPLES OF METALS & APPLICATIONS

- Traditional Ceramics:传统陶瓷:
 - Brick, porcelain, glass 砖、瓷器、玻璃
- Building materials (Tiles, Bricks, Cement) 瓷砖、砖块、水泥
- Heat-Resistant Parts (engine components) 耐热部件(发动机部件)
- Electrical Insulators 电绝缘体
- Bioceramics (dental implants, artificial bones) 生物陶瓷(牙科植入物、人造骨骼)

GENERAL CHARACTERISTICS OF CERAMICS

- **Appearance 外观**
 - Often dull暗淡, opaque不透明, or white呈白色
 - Some can be translucent半透明性 (like glass or alumina氧化铝)
- **Hardness and Strength 硬度和强度**
 - Super hard and strong under compression 在受压状态下异常坚固且强劲
 - Do not deform plastically and fracture suddenly不会塑性变形会突然断裂
 - Brittle (breaks easily under tension) 脆性(在受拉力或冲击时容易破裂)
- **Density 密度**
 - Generally lower than metals but higher than polymers
- **Electrical Conductivity 导电性:**
 - Electrical insulators 大多数陶瓷材料是绝缘体
- **Thermal Conductivity 导热性:**
 - Moderate thermal conductivity 中等导热性
 - Poor thermal shock resistance(can crack when rapidly change on Temp)
热震性能较差
- **Melting and Boiling Points 熔点和沸点**
 - Excellent heat resistance出色的耐热性 -> High melting points高熔点
- **Chemical Resistance 化学耐性**
 - Highly resistant to corrosion, oxidation极强抗腐蚀、抗氧化

知识点3: Polymers高分子聚合物

Polymers are **long-chain molecules** made of repeating structural units (monomers). They are mostly **organic** 有机 (**carbon-based**) materials and include plastics, rubbers, and fibers 塑料、橡胶和纤维

Examples of Natural and Synthetic Polymers

Natural polymers are made by living organisms.



DNA



Rubber



Cellulose



Wool

Synthetic polymers are made by chemical reactions in a lab.



Nylon



Polyester



Teflon



Epoxy

sciencenotes.org

EXAMPLES OF POLYMERS & APPLICATIONS

- Natural Polymers: Rubber, proteins
天然聚合物: 橡胶、纤维素、蚕丝、蛋白质
- Synthetic Polymers: Polyethylene (PE), Polyvinyl chloride (PVC), Nylon, Teflon
人工合成聚合物: 聚乙烯(PE)、聚氯乙烯(PVC)、尼龙、特氟龙
- Packaging (plastic bags, bottles) 包装(塑料袋、瓶子)
- Textiles (nylon, polyester) 纺织品(尼龙、聚酯纤维)
- Insulators (wires, electronics) 绝缘体(电线、电子元件)
- Medical (contact lenses, prosthetics) 医疗(隐形眼镜、假肢)
- Everyday plastics (containers, toys) 日常塑料制品(容器、玩具)

GENERAL CHARACTERISTICS OF POLYMERS

- **Appearance 外观**
 - Transparent, Translucent, or Colored 透明的、半透明的或者彩色的
 - **Strength & Flexibility 强度与柔韧性**
 - Generally low strength 通常强度较低
 - flexible and elastic 灵活且有弹性的
 - **Durability 持久与耐用性**
 - Can degrade over time 可能会随着时间的推移而发生降解
 - Become brittle under Sunlight 塑料在阳光照射下会变得变脆
 - **Density 密度**
 - Low density → very lightweight compared to metals and ceramics
 - **Electrical Conductivity 导电性:**
 - Poor conductors (good electrical insulators)(良好的电绝缘体)
 - **Thermal Conductivity 导热性:**
 - Low melting/softening points → poor thermal conductivity
 - **Chemical Resistance 化学耐性**
 - Resistant to many chemicals, but some degrade under UV light or solvents
- 对许多化学物质具有抗性, 但有些会在紫外线照射下或在溶剂作用下发生分

解

本节知识点总结

Comparison of Material Classes

Property	Metals 🏗️	Ceramics 🪨	Polymers 🌿
Bonding	Metallic bonding ("sea of electrons")	Ionic & covalent bonds	Covalent bonds within chains; Van der Waals between chains
Appearance	Shiny (metallic luster)	Dull, often opaque or translucent	Dull, can be transparent or colored
Density	High	Medium to high	Low
Strength	High (can deform before breaking)	Very high compressive strength, brittle	Low to moderate, flexible
Electrical Conductivity	Excellent (free electrons)	Poor (insulators)	Poor (insulators) unless doped
Thermal Conductivity	High	Low to medium	Low
Malleability & Ductility	Good (can be shaped, drawn)	Poor (brittle, shatter easily)	Good (flexible, elastic)
Melting Point	Generally high	Very high	Low to moderate
Examples	Iron, Copper, Aluminum	Glass, Porcelain, Alumina (Al_2O_3)	Nylon, PVC, Polyethylene
Applications	Construction, wiring, vehicles	Tiles, engine parts, space shuttles	Packaging, textiles, plastics



练习题

Which of the following are examples of Metals, Ceramics, and Polymers?

1. An aluminum beverage container
2. Paper
3. A gold wedding ring
4. The bottom of a running shoe
5. A wooden table
6. A conventional glass window
7. The flexible covering on a computer power cord
8. A brick wall
9. The copper conductor in a computer power cord
10. The mortar that holds together bricks in a brick wall 砖块之间的泥浆
11. A candle
12. The lining of a high temperature furnace 高温熔炉的内衬
13. The frame of a conventional automobile
14. The tires on a car
15. The wheels on a car
16. A semiconductor
17. A carbon fibre reinforced polymer aircraft component



Metals

- 1). **An aluminum beverage container** → Metal (Aluminum)
- 3). **A gold wedding ring** → Metal (Gold)
- 9). **The copper conductor in a computer power cord** → Metal (Copper)
- 13). **The frame of a conventional automobile** → Metal (Steel, usually Iron alloy)
- 15). **The wheels on a car** → Metal (often aluminum alloys or steel)

Ceramics

- 6). **A conventional glass window** → Ceramic (Glass = non-crystalline ceramic)
- 8). **A brick wall** → Ceramic (Brick = fired clay ceramic)
- 10). **The mortar that holds together bricks in a brick wall** → Ceramic (lime石灰/silica/calcium-based material)
- 12). **The lining of a high temperature furnace** → Ceramic (refractory materials, e.g., alumina, silica)

Polymers

- 4). **The bottom of a running shoe** → Polymer (rubber or synthetic polymeric material)
- 7). **The flexible covering on a computer power cord** → Polymer (plastic insulation, usually PVC or polyethylene)
- 11). **A candle** → Polymer (paraffin wax石蜡[常用于密封材料] = hydrocarbon polymer-like material)
- 14). **The tires on a car** → Polymer (rubber, synthetic)

下节知识点

	知识点	考法	难度
1	Define Hook's Law, Engineering Stress, and Engineering Strain	概念/计算	★★★★★
2	Describe Conventional Uniaxial Tensile Test	概念	★★★★
3	Identify typical "dog-bone" tensile specimen: reduced sections, gauge length, cross sectional area, grip region	概念	★★
4	Describe elastic deformation in term of macroscopic behaviour and atomic positions	概念	★★
5	Generalized interatomic force v.s. Separation curve related to Young's Modulus	看图/记忆	★★★

